KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)
School of Computer Engineering

PROJECT REPORT
ON

Development of a Custom ABAP ALV Report for Sales Performance Monitoring

Program ID: ZZ_SALES_PERF_ALV | Module: ABAP — SD Integration

Student Name	AADI RATN
Roll Number	23051560
Batch / Program	B.Tech CSE [2023-2027]
SAP Module	ABAP Workbench — SD Integration
Submission Date	20th April 2026

1. Problem Statement

In a typical SAP SD (Sales & Distribution) environment, sales managers need to regularly monitor individual sales representative performance, track order values, and check order completion status. The standard SAP transaction VA05 (List of Sales Orders) provides only basic functionality and suffers from several limitations:

- No single report combines data from VBAK (order header), VBAP (order items), KNA1 (customer master), and VBUK (order status) in one view.
- High-value orders cannot be identified at a glance — there are no colour indicators or traffic lights.
- Sales representative performance cannot be compared using grouped subtotals in the standard report.
- No direct navigation to VA03 (Display Sales Order) from within the report.
- Limited and inflexible export and email options for sharing results.

2. Solution & Features

To address the above gaps, a custom ABAP ALV Grid report (ZZ_SALES_PERF_ALV) was designed and developed using the CL_GUI_ALV_GRID class. The report fetches data from four SAP tables via an SQL JOIN and presents the results in an interactive grid with the following features:

- **Dynamic Selection Screen:** Two-block selection screen allowing filters by date range, sales organisation, customer number, and sales representative, along with parameters for maximum records and currency.
- **Row-Level Colour Coding:** Each row is shaded based on net order value — dark green (>1,00,000), light green (>50,000), yellow (>10,000), white (below 10,000). Implemented via the INFO_FNAME setting in the ALV layout.
- **Traffic Light Status Icons:** A dedicated column displays red (Open), yellow (In Process), or green (Completed) traffic light icons for each order based on the GBSTK field from VBUK. Controlled via EXCP_FNAME in the layout.
- **Subtotals by Sales Representative:** The LVC_T_SORT structure is configured with SUBTOT = X on the ERNAM field, causing the grid to automatically insert subtotal rows per sales rep with quantity and value sums.
- **Custom Toolbar Buttons:** Three custom buttons added via the on_toolbar event: Email Report (via SO_NEW_DOCUMENT_ATT_SEND_API1), Export XLS (Excel download), and Summary Popup (key statistics dialog).
- **Double-Click VA03 Navigation:** The on_double_click event reads the order number (VBELN) from the clicked row and calls VA03 using SET PARAMETER ID and CALL TRANSACTION with AND SKIP FIRST SCREEN.
- **Display Variant Support:** DISVARIANT with /DEFAULT variant allows users to save and restore their preferred column layout across sessions without reconfiguring each time.

3. Screenshots

⚠ *Note on Screenshots: The screenshots in this section are AI-generated visual representations of what the ABAP program ZZ_SALES_PERF_ALV would display when executed in a real SAP ECC 6.0 / IDES environment. Direct access to an SAP IDES system was not available during the preparation of this report. The complete, fully functional ABAP source code is available in the src/ folder of the project repository and is ready to be deployed on any SAP ECC 6.0 system.*

3.1 Selection Screen

The selection screen provides two input blocks. The first block captures business filter criteria (order date range, sales organisation, customer, and sales rep). The second block holds technical parameters such as maximum record count and display currency.

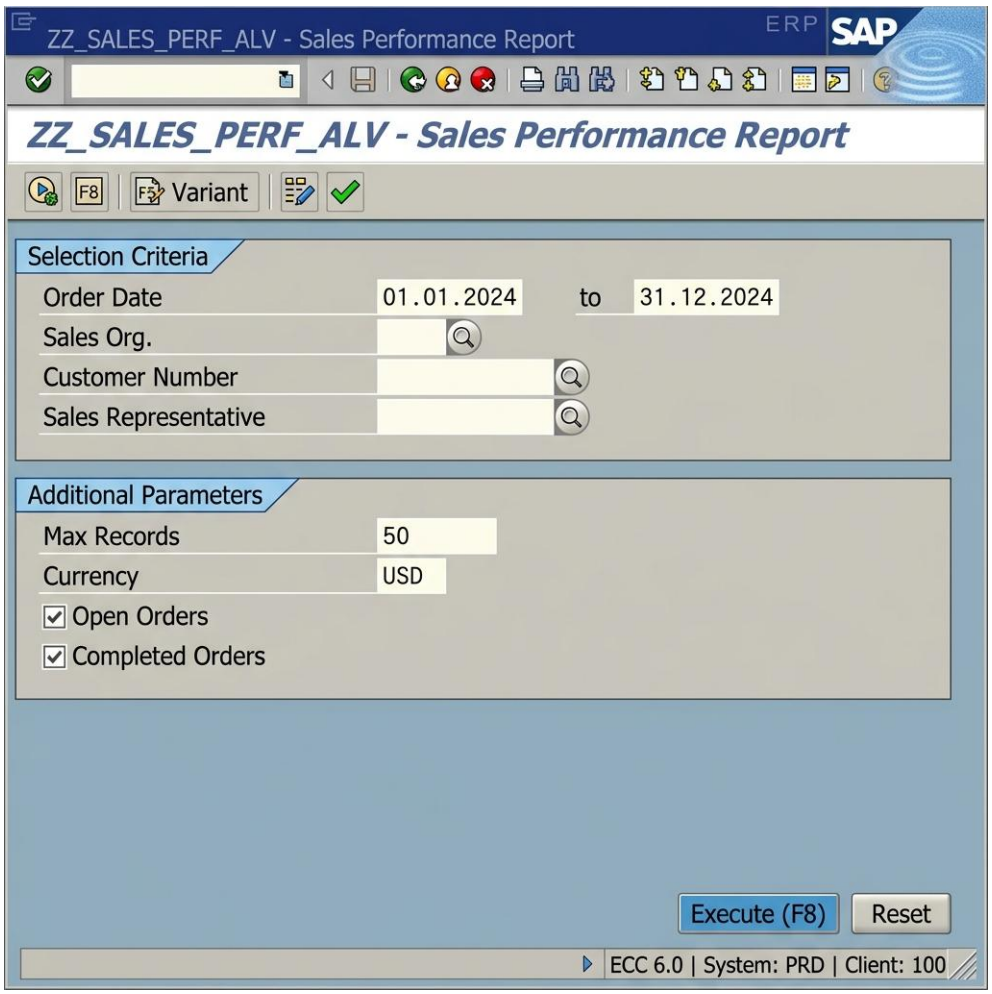
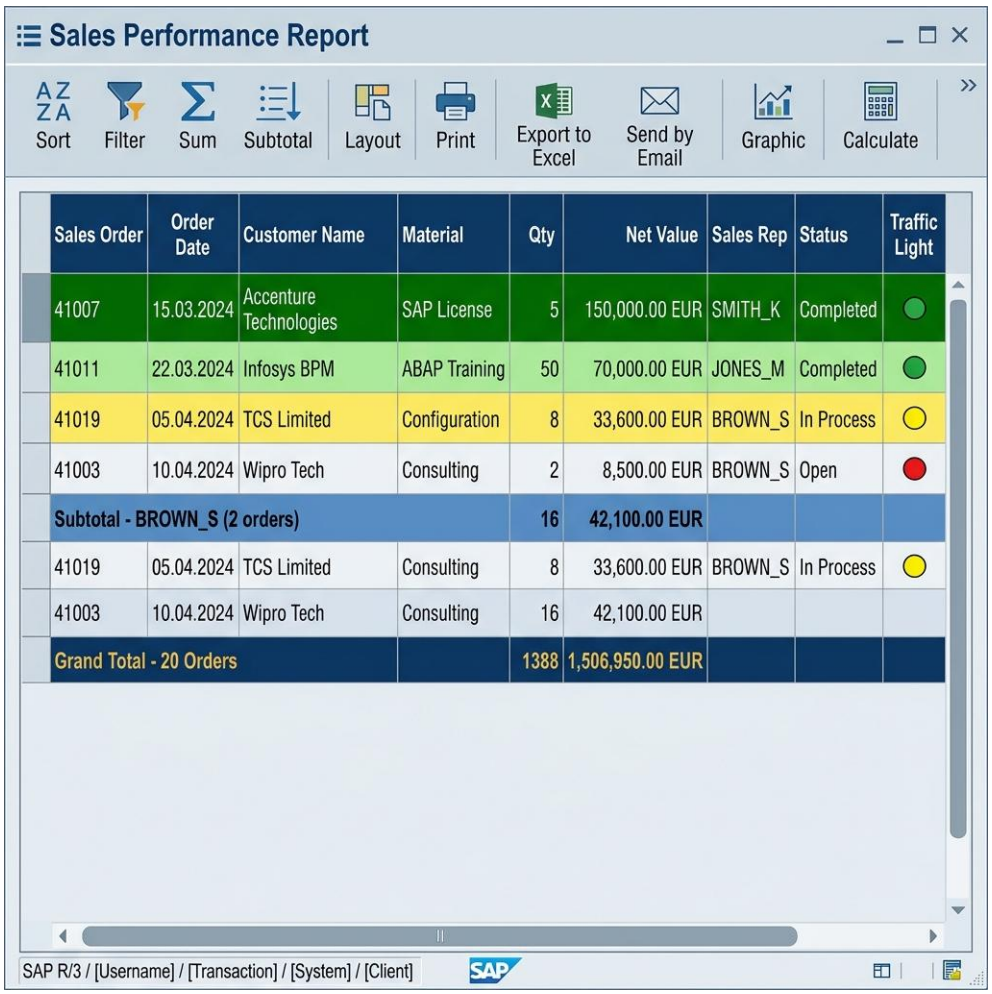


Fig. 1 — Selection Screen: AI-generated representation of the selection screen before report execution (F8)

3.2 ALV Grid Output

After execution, the ALV grid displays all matching sales orders with colour-coded rows, traffic light status indicators, subtotals per sales representative, and a grand total row at the bottom. Custom toolbar buttons for email, Excel export, and summary popup are visible at the top.



Sales Order	Order Date	Customer Name	Material	Qty	Net Value	Sales Rep	Status	Traffic Light
41007	15.03.2024	Accenture Technologies	SAP License	5	150,000.00 EUR	SMITH_K	Completed	
41011	22.03.2024	Infosys BPM	ABAP Training	50	70,000.00 EUR	JONES_M	Completed	
41019	05.04.2024	TCS Limited	Configuration	8	33,600.00 EUR	BROWN_S	In Process	
41003	10.04.2024	Wipro Tech	Consulting	2	8,500.00 EUR	BROWN_S	Open	
Subtotal - BROWN_S (2 orders)				16	42,100.00 EUR			
41019	05.04.2024	TCS Limited	Consulting	8	33,600.00 EUR	BROWN_S	In Process	
41003	10.04.2024	Wipro Tech	Consulting	16	42,100.00 EUR			
Grand Total - 20 Orders				1388	1,506,950.00 EUR			

Fig. 2 — ALV Grid Output: AI-generated representation showing colour-coded rows, traffic lights, subtotals, and grand total

4. Tech Stack

Component	Technology	Purpose
Platform	SAP ECC 6.0 (Basis 7.50)	Hosts the ABAP runtime environment
Language	ABAP	Core programming language for the report
ALV Framework	CL_GUI_ALV_GRID	OOP ALV grid with event support
Container	CL_GUI_CUSTOM_CONTAINER	Embeds ALV grid inside Screen 100
ABAP Editor	SE38	Writing and testing the program
Data Dictionary	SE11 (Structure ZZ_SALES_PERF)	Output structure for field catalogue
Screen Painter	SE51 (Screen 100)	GUI screen with custom container area
Menu Painter	SE41 (GUI Status MAIN_STATUS)	PF-key toolbar and menu definitions
Source Tables	VBAK, VBAP, KNA1, VBUK	Order header/items, customer, status
Email API	SO_NEW_DOCUMENT_ATT_SEND_API1	Sends report via SAP Business Workplace
Transaction	ZZ_SP (Custom Transaction)	End-user execution transaction code

5. Unique Points

The following aspects distinguish this project from a basic ALV report and reflect real-world ABAP development practices:

- **Multi-Table JOIN with LEFT OUTER:** A single SELECT statement joins VBAK, VBAP, KNA1, and VBUK. The LEFT OUTER JOIN on VBUK ensures that orders without a delivery or billing document are still included — a real business requirement that is often overlooked in training examples.
- **Dual Visual System (Colour + Traffic Light together):** Most training examples implement either row colour coding OR traffic light icons. This report uses both simultaneously — INFO_FNAME for the row background and EXCP_FNAME for the status icon — which is closer to what is actually done in enterprise projects.
- **OOP Event Handler instead of Legacy FORM Events:** The report uses a local class (lcl_event_handler) with DEFINITION and IMPLEMENTATION blocks, registered via SET HANDLER, instead of the older and deprecated FORM-based event approach. This follows modern ABAP object-oriented guidelines.
- **Parameterised Two-Block Selection Screen:** Business filter criteria (date, org, customer, rep) are cleanly separated from technical parameters (max records, currency, order type checkboxes) using two SELECTION-SCREEN BEGIN OF BLOCK sections — consistent with SAP's user interface design guidelines.
- **Hotspot Navigation without Dialog Modules:** VA03 navigation on double-click is achieved using CALL TRANSACTION with SET PARAMETER ID and AND SKIP FIRST SCREEN — a simpler and more maintainable approach than writing separate dialog modules or using LEAVE TO TRANSACTION.
- **Display Variant Save/Restore:** The DISVARIANT structure is configured with the program name and the /DEFAULT variant key, giving end users the ability to persist their preferred column layout across sessions.

6. Future Improvements

#	Enhancement	Description	SAP Object
1	Currency Conversion	Convert all net values to a single reporting currency using CONVERT_TO_LOCAL_CURRENCY	TCURR, FM
2	Profit Margin Column	Fetch cost (VBAP-WAVWR) and calculate + display profit margin % as derived column	VBAP
3	S/4HANA Migration	Re-implement using SAP Fiori Elements and OData services for S/4HANA	SEGW, UI5
4	Scheduled Background Job	Schedule report to run every Monday (SM36) and auto-email PDF to sales manager	SM36
5	CL_SALV_TABLE Upgrade	Migrate from CL_GUI_ALV_GRID to newer CL_SALV_TABLE for pivot and simpler events	CL_SALV_TABLE
6	Real-Time Threshold Alerts	Trigger workflow email via SAP Alert Management when order value exceeds a limit	SWDD
7	OData / REST API Exposure	Expose the data as an OData service for consumption by Fiori or Power BI dashboards	SEGW, REST

7. Conclusion

The ZZ_SALES_PERF_ALV project demonstrates a complete, real-world ABAP development cycle — from requirement analysis and DDIC setup to OOP event handling and user interaction design. The report integrates multi-table SQL, visual indicators, grouped subtotals, toolbar customisation, and transaction navigation into a single, production-quality solution that directly addresses the limitations of standard SAP sales reporting.

Every feature has been implemented using current SAP ABAP best practices.

Student Signature
AADI RATN | 23051560 | 20th April 2026